Learning Where to Propagate and What to Emphasize: A Unified Spatio-Temporal Graph Architecture for Human-Centric Data

Supplementary Material

1 Experimental Setup

Facial Expression Recognition: Experiments are conducted on AFEW 8.0, Oulu-VL, CAS(ME)^3, and SMIC-HS. AFEW follows the official training and validation splits, while Oulu-VL uses subject-independent 10-fold cross-validation. CAS(ME)^3 and SMIC-HS are evaluated using leave-one-subject-out (LOSO) cross-validation. Cross-dataset evaluation is performed between Oulu-VL and CK+ under the six-class setting, and between SMIC-HS and CASME II under the three-class setting, in both training-to-testing directions. Facial frames are resized to 224×224 pixels, and 51 inner-face landmarks are retained from the 68 detected landmarks. Dense Farnebäck optical flow is computed between selected frames, and 7×7 magnitude patches are extracted around each landmark. The temporal length is $T = 8$ optical-flow intervals computed from nine frames, except for CAS(ME)^3, where $T = 6$.

Epileptic Seizure Detection: Experiments are conducted on CHB-MIT and Siena using cross-patient leave-one-patient-out (LOPO) evaluation. CHB-MIT uses 18 bipolar EEG channels, while Siena uses 16 bipolar channels and is resampled to 256 Hz. EEG preprocessing includes 0.5–70 Hz band-pass filtering, 50/60 Hz notch filtering, FastICA-based artifact removal, and z-score normalization. Recordings are divided into 4-second segments, with 75% overlap for seizure segments and no overlap for non-seizure segments. Each segment is further divided into seven overlapping 1-second windows with a 0.5-second step. Class balancing is applied only to training data, preserving the original distribution of held-out patients.

Human Action Recognition: Experiments are performed on NTU RGB+D 60 and NTU RGB+D 120 using their official evaluation protocols. NTU 60 employs cross-subject (X-Sub) and cross-view (X-View) evaluation, while NTU 120 uses cross-subject (X-Sub) and cross-setup (X-Set) evaluation. Each skeleton sequence is temporally normalized to $T = 64$ frames with $N = 25$ joints. Joint coordinates are normalized relative to a reference joint, and bone features are computed from parent–child joint displacements. Two independently trained joint and bone streams are used, with their class probabilities averaged during inference.

2 Implementation Details

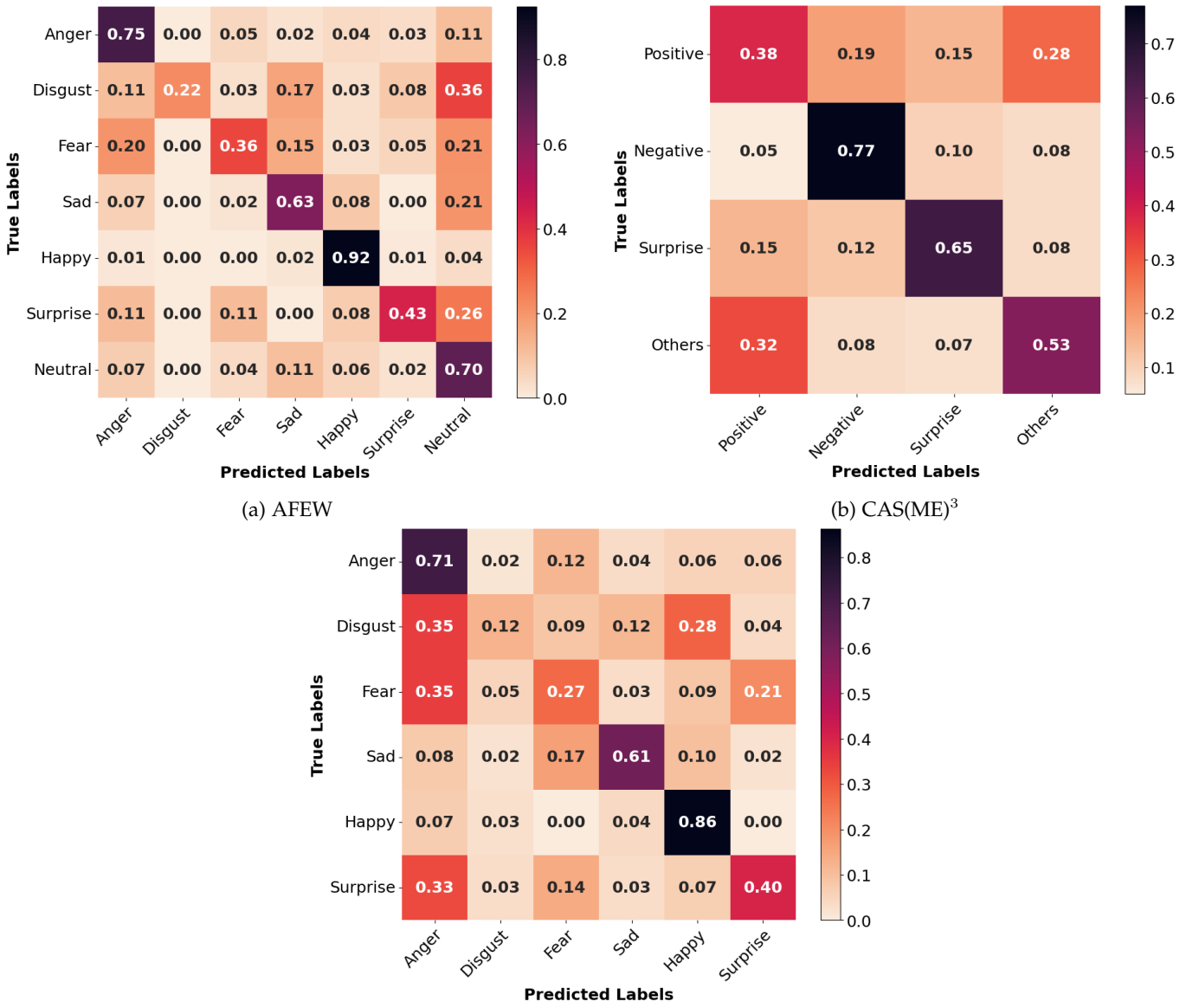
All experiments are implemented in PyTorch using a single NVIDIA RTX 3090 GPU with 24 GB of memory. The proposed framework maintains the same architectural components across tasks, with modality-specific input representations and independent training.

Network Configuration: Input node features are projected into a common 64-dimensional space. The Structural Encoder contains two ST-DAGCN blocks with channel dimensions $(64, 128)$ and an ST-DA reduction ratio of 4. The Contextual Encoder uses a 128-dimensional representation, Hydra with state dimension 16, and spatial self-attention with dimension 32. The Gated Decoder employs a relation dimension of 32, temporal decoding radius $\delta = 1$, and hidden dimensions of 64 for both gating and temporal-pair scoring. ST-TAGCN contains three blocks with channels $(64, 128, 256)$. The temporal kernel size is 3, and the modulation radius is $r_{\mathrm{mod}} = 1$. Spatial and temporal convolutions employ the separable SGLayer bottleneck [1] with a reduction ratio of 2.

Training Configuration: For facial expression recognition, Adam is used with an initial learning rate of 10^{-3}, weight decay of 10^{-4}, cosine annealing, batch size 32, and 100 epochs. Dropout is set to 0.3, and data augmentation includes random scaling, horizontal flipping, and rotation. For seizure detection, Adam is employed with a learning rate of 10^{-4}, batch size 128, dropout rate 0.2, and 100 epochs. For action recognition, SGD is used with momentum 0.9, weight decay 4×10^{-4}, and an initial learning rate of 0.025. A five-epoch warmup is applied, followed by tenfold learning-rate reductions at epochs 110 and 120. Training continues for 140 epochs with batch size 64. All models are optimized using cross-entropy loss.

2.1 Ablation Study

We conduct ablation studies to examine the effectiveness of the proposed components across the three tasks. For CAS(ME)^3, we report the four-class evaluation protocol. We additionally report results on the Mixed dataset for facial expression recognition, which combines AFEW, Oulu-VL, CAS(ME)^3, and SMIC-HS for joint macro- and micro-



(a) AFEW

(b) CAS(ME)^3

(c) Mixed

Fig. S1: Confusion matrices of the proposed framework for macro- and micro-expression recognition on the AFEW, CAS(ME)^3, and Mixed datasets.

TABLE S1: Effect of the number of selected keyframes on macro- and micro-expression recognition performance across three datasets.

Number of Keyframes	**AFEW** *(WAR %)*	**CAS(ME)^3** *(UAR %)*	**Mixed** *(WAR %)*
4	59.54	78.30	54.48
6	62.51	**79.25**	59.76
8	**63.11**	79.16	**65.29**
10	61.44	74.91	64.63
12	57.13	72.72	63.86

expression recognition under a unified six-class emotion setting.

Figure S1 presents the normalized confusion matrices of the proposed method on AFEW, CAS(ME)^3, and Mixed for expression recognition. On AFEW, the model shows strong recognition of Happy, Anger, Neutral, and Sad, with class-wise recalls of 0.93, 0.76, 0.70, and 0.63, respectively, while Disgust and Fear remain more challenging and are frequently confused with Neutral and Anger. On CAS(ME)^3, the predictions are strongly concentrated along the diagonal, achieving recalls of 0.90 for Negative, 0.88 for Surprise, and 0.77 for Others, whereas Positive is more frequently confused with Others. A similar trend is observed on the Mixed dataset, where Happiness and Anger obtain the highest recalls of 0.93 and 0.86, followed by Sad at 0.74, while Fear, Disgust, and Surprise exhibit larger inter-class confusion, particularly toward Anger. Overall, the confusion matrices indicate that the proposed framework effectively distinguishes visually distinctive expression categories, while most remaining errors occur among classes with more subtle or overlapping facial dynamics.

Table S1 investigates the effect of the number of

keyframes on facial expression recognition performance. Increasing the number of keyframes from 4 to 8 generally improves the results, indicating that a moderate number of frames provides more informative temporal coverage of the expression sequence. The Mixed dataset, which combines MaE and ME samples from multiple sources, is also evaluated to assess the effect of keyframe selection under more heterogeneous expression conditions. On AFEW and Mixed, the best performance is obtained with 8 keyframes, achieving WARs of 63.11% and 65.29%, respectively. On CAS(ME)^3, 6 keyframes yield the highest UAR of 79.25%, while 8 keyframes achieve a comparable 79.16%. Further increasing the number of keyframes to 10 or 12 degrades performance across the datasets, suggesting that excessive temporal sampling may introduce redundant or less informative frames. Accordingly, we use 8 keyframes for AFEW and Mixed and 6 keyframes for CAS(ME)^3, allowing the temporal sampling strategy to better match the characteristics of each dataset.

REFERENCES

[1] Y.-F. Song, Z. Zhang, C. Shan, and L. Wang, "Constructing stronger and faster baselines for skeleton-based action recognition," *IEEE transactions on pattern analysis and machine intelligence*, vol. 45, no. 2, pp. 1474–1488, 2022.